Module 6
Virtualization Types

Server Virtualization

Server Virtualization

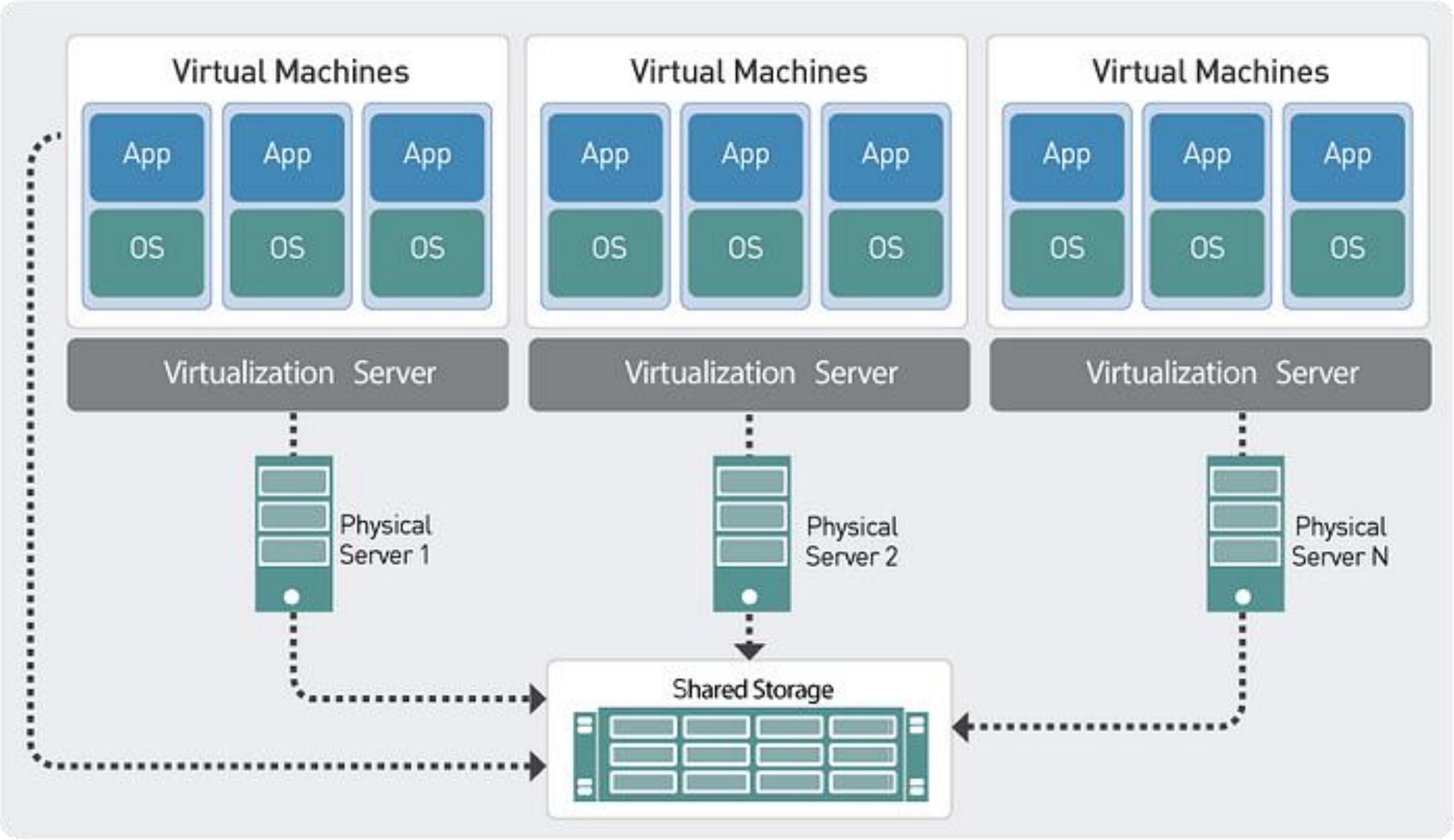
Server virtualization is the *process of creating multiple server instances from one physical server*.

Each server instance represents an isolated virtual environment. Within each virtual environment, they can run a separate operating system.

It is an *efficient and cost-effective* way to use server resources and deploy IT services in an organization. Without server virtualization, physical servers use only a small amount of their processing capacities, which leave devices idle.

Server virtualization abstracts server software from hardware on a guest/host basis, enabling multiple virtual servers to run on a physical device.

Server Virtualization



Why Server Virtualization?

On average, dedicated servers use only 15% of their resources during normal operation - it is a waste of resources.

Software or hardware failures often required hands-on repair on all the servers.

Therefore, there was a *need to boost resource utilization and maintain separation between the clients' operating systems for security purposes.*

It allows the organizations to cut down on the number of physical servers, saving money and reproducing hardware footprints.

Server Virtualization

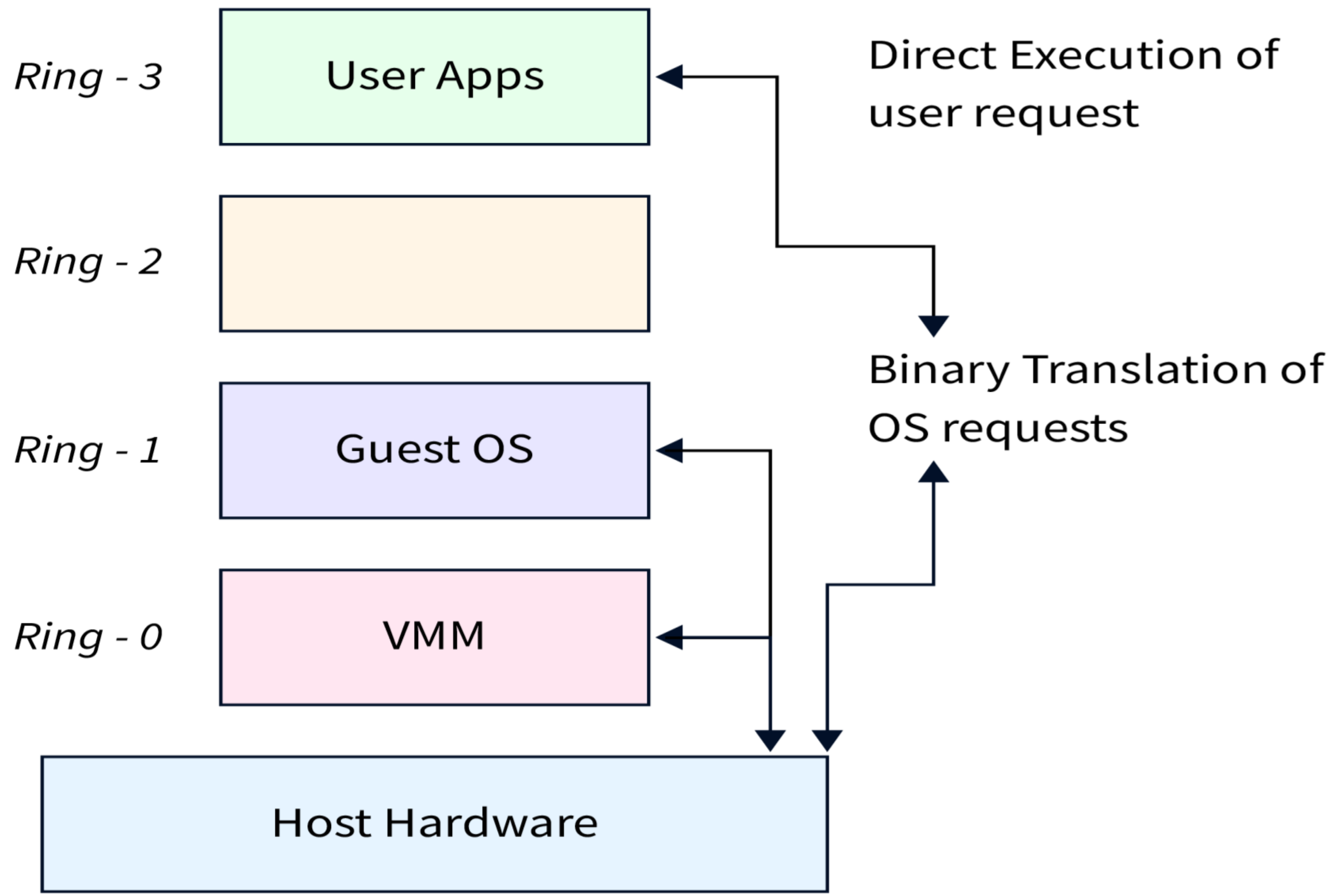
- To create virtual server instances you first need to set up a virtualization software called Hypervisor.
- Its main role is to create a virtualization layer that separates CPU / Processors, RAM and other physical resources from the virtual instances.
- Once you install the hypervisor on your host machine, you can use that virtualization software to emulate the physical resources and create a new virtual server on top of it.

Server virtualization can be implemented using

1. Full virtualization
2. Para-virtualization
3. OS level - virtualization

The distinction between these three approached are mainly based on the level of isolation they provide, which is also related to how much hardware resources they emulate.

Full Virtualization



Full Virtualization

- It was developed by ***IBM*** in ***1966***.
- A type 1 hypervisor, also referred to as a native or bare metal hypervisor, runs directly on the host's hardware to manage guest operating systems.
- It takes the place of a host operating system and VM resources are scheduled directly to the hardware by the hypervisor.
- It operates by combining binary translation and direct compilation, where the guest operating system is completely separated from the basic hardware and virtualization layer.
- Full virtualization relies on a hypervisor and uses hardware-assisted virtualization technologies to run guest OS directly on the physical CPU.

Full Virtualization

- VMs created through full virtualization are completely isolated from each other.
- Each VM runs its own instance of the guest operating system, which cannot interfere with other VMs.
- When an OS instruction is created, the hypervisor immediately translates it during run-time and stores the result for future reference. At the same time, the user-level instructions are run without changes at native speed.
- This type of hypervisor is most common in an enterprise data center or other server-based environments.
- Popular hypervisors for full virtualization include **VMware vSphere/ESXi**, **Microsoft Hyper-V**, and **Oracle VirtualBox**.

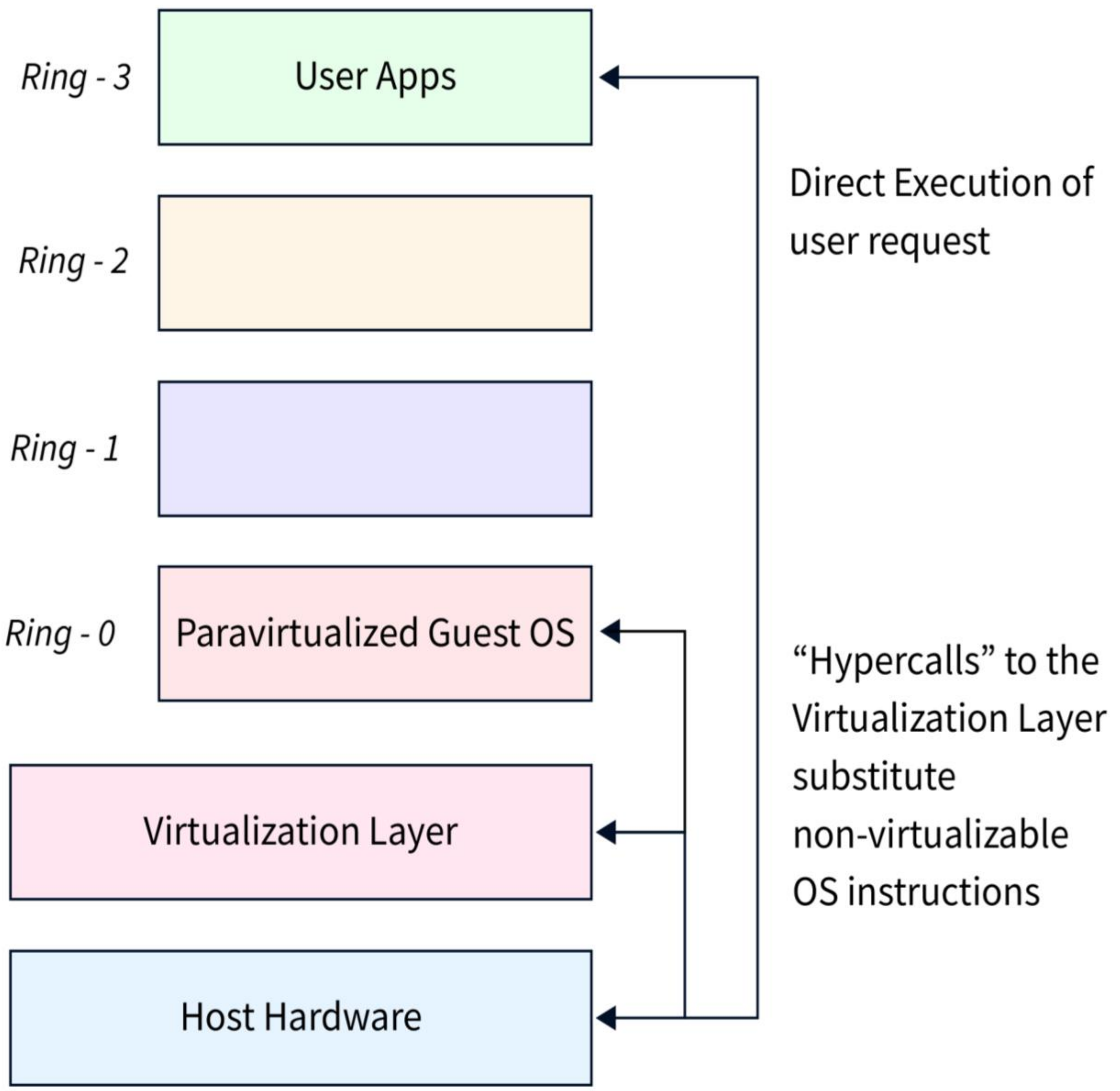
Para Virtualization

- A type 2 hypervisor is also known as a hosted hypervisor, and is run on a conventional operating system as a software layer or application.
- It works by abstracting guest operating systems from the host operating system. VM resources are scheduled against a host operating system, which is then executed against the hardware.
- Paravirtualization is a type of virtualization where software instructions from the guest operating system running inside a virtual machine can use "hypercalls" that communicate directly with the hypervisor.
- Here, the guest OS is not completely isolated but it is partially isolated by the virtual machine from the virtualization layer and hardware.

Para Virtualization

- It also alters the operating system kernel to use hypercalls rather than non-virtualizable instructions. The goal of hypercalls is to communicate with the virtualization layer hypervisor directly.
- Paravirtualization replaces nonvirtualizable instructions with hypercalls that communicate directly with the virtualization layer hypervisor.
- Paravirtualization requires guest OS to use a specific set of APIs to interact with the virtualization layer.
- A type 2 hypervisor is better for individual users who want to run multiple operating systems on a personal computer.
- Xen is a widely-used hypervisor that supports para-virtualization.

Para Virtualization



How does Server Virtualization work?

Server virtualization lets different operating systems and applications operate on one physical server.

Hypervisor: The hypervisor is the software layer that powers server virtualization. It separates the server's hardware from its virtual computers and assigns server cores, memory, storage, and networking to virtual machines.
Virtual Machines (VMs): Virtual machines are software simulations of actual computers. The operating system and applications operate separately on the same physical server as the other VMs. Each VM uses the hypervisor's virtualized hardware resources like a physical server.
Resource allocation: The hypervisor dynamically distributes physical resources to each VM. In case one VM needs more CPU and another needs more RAM, the hypervisor can adjust the resource allocation.

How does Server Virtualization work?

Server virtualization lets different operating systems and applications operate on one physical server.

Isolation: The virtual machines are segregated, so if one crashes or faces troubles, it doesn't impact the others. The hypervisor maintains VM connectivity and secures their independence.

Hardware utilization: Server virtualization optimizes server hardware. Virtual machines can run several operating systems and applications on the same hardware, optimizing resource usage and decreasing the need for actual servers.

Server Virtualization

Benefits:

- Cost savings, High flexibility and scalability
- Better server utilization
- Enhanced disaster recovery and business continuity
- Isolation and Security
- Reduced energy consumption
- Support for legacy applications

Server Virtualization

Disadvantage:

- Overheads
- Resource contention
- Single point of failure
- Complexity and Security risks
- Software licensing needs
- Need for experienced IT staff

Storage Virtualization

Storage Virtualization

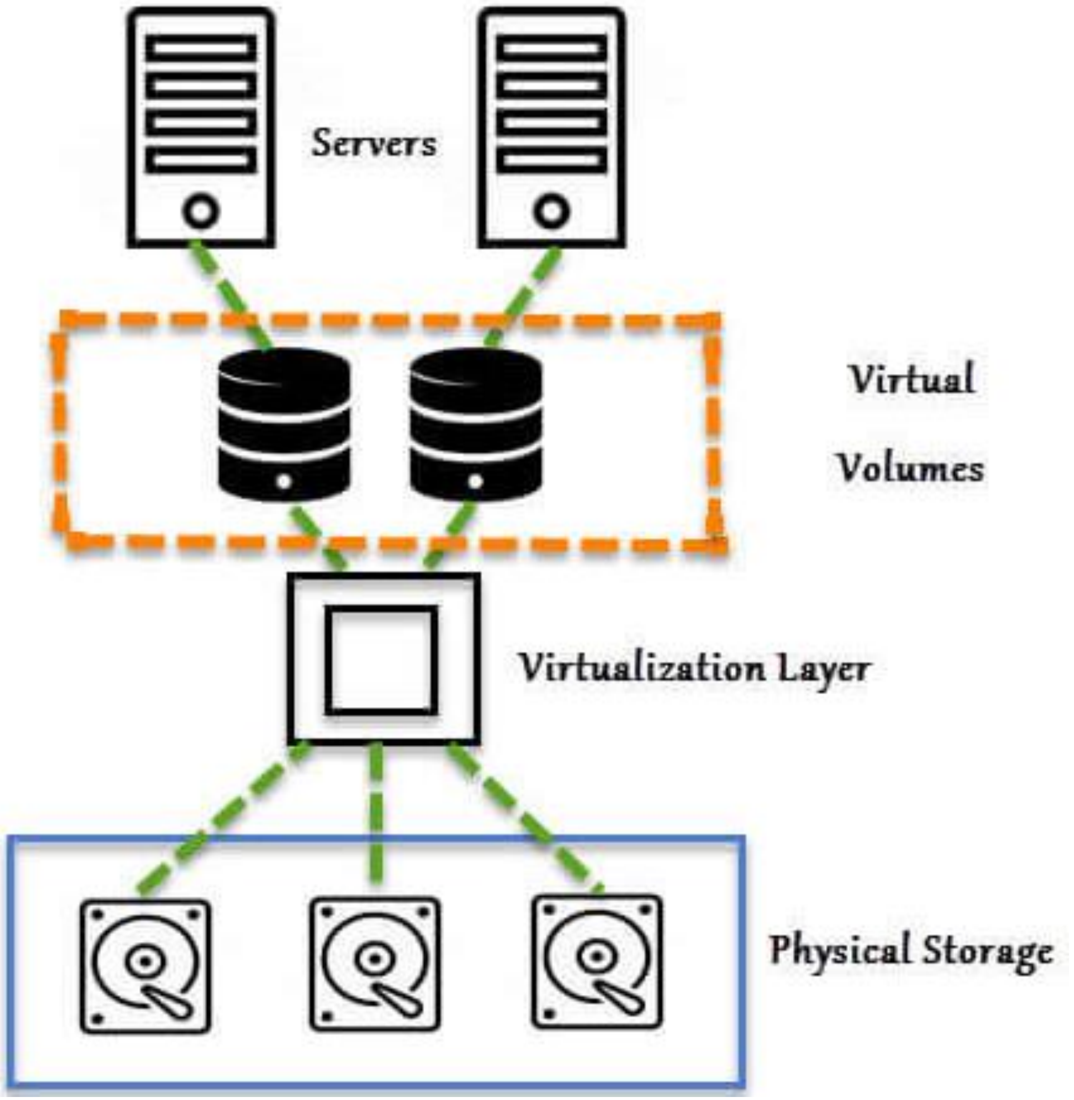
Storage virtualization is the process of abstracting physical storage resources and presenting them as logical storage units to be used by applications or other systems.

It allows for *better utilization*, *management*, *and flexibility* of storage infrastructure.

It is used in data centers of organizations where there is a *huge requirement for storing a large amount of data*.

Storage Virtualization

virtual storage pool



How it works?

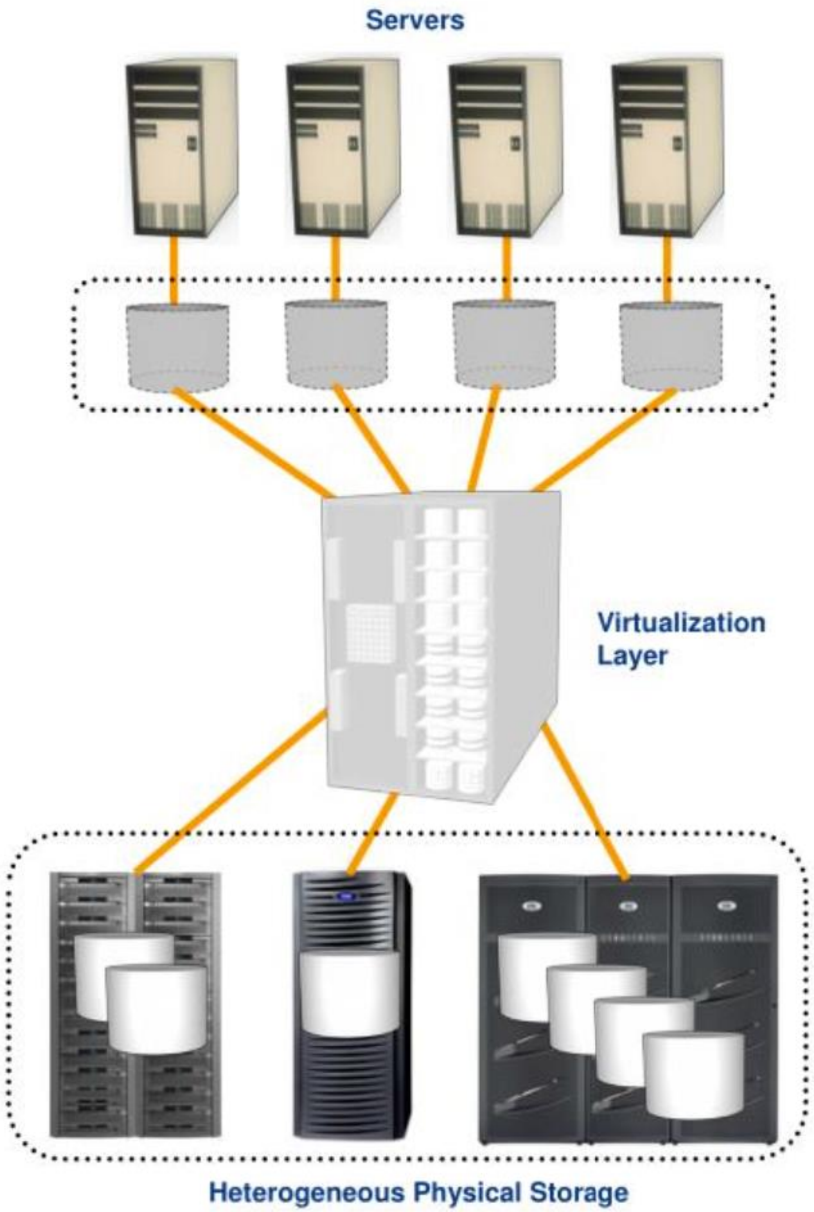
Virtualization Layer:

Storage virtualization is facilitated by a virtualization layer, which can be implemented as software or hardware. This layer acts as an *intermediary between the physical storage infrastructure and the systems accessing the storage.*

Storage Pool Creation:

The virtualization layer aggregates and combines the available physical storage resources from various storage devices or arrays into a virtual storage pool. This pool can *span multiple storage devices, types, and vendors*, creating a unified and flexible storage infrastructure.

Storage Virtualization



How it works?

Provisioning of Virtual Volumes:

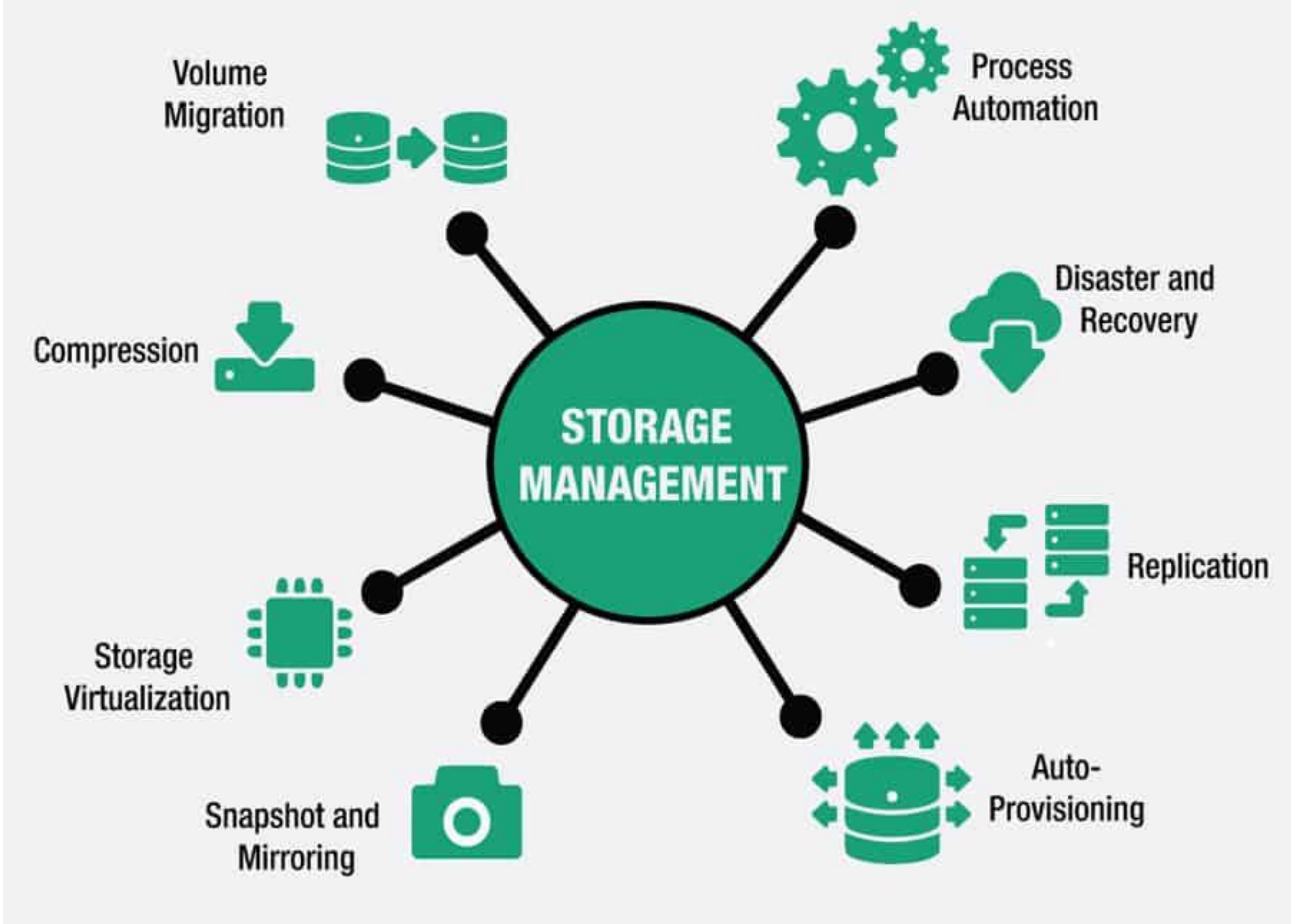
From the virtual storage pool, virtual volumes are created and presented to the systems or applications requiring storage. These *virtual volumes appear as logical units* and can be provisioned to meet specific capacity and performance requirements.

Abstraction and Mapping:

The virtualization layer handles the abstraction and mapping of the logical storage units to the physical storage resources. It *maintains a mapping table* that tracks which logical volumes correspond to which physical storage devices or arrays.

Storage Management and Services:

The virtualization layer provides management capabilities and services for the virtual storage pool. This includes functions like *storage provisioning, data protection (such as snapshots and replication), performance optimization, and monitoring*.



- Centralized Administration
- *capacity allocation, performance optimization, and data protection*
- flexibility and scalability by decoupling the logical storage units from the physical hard

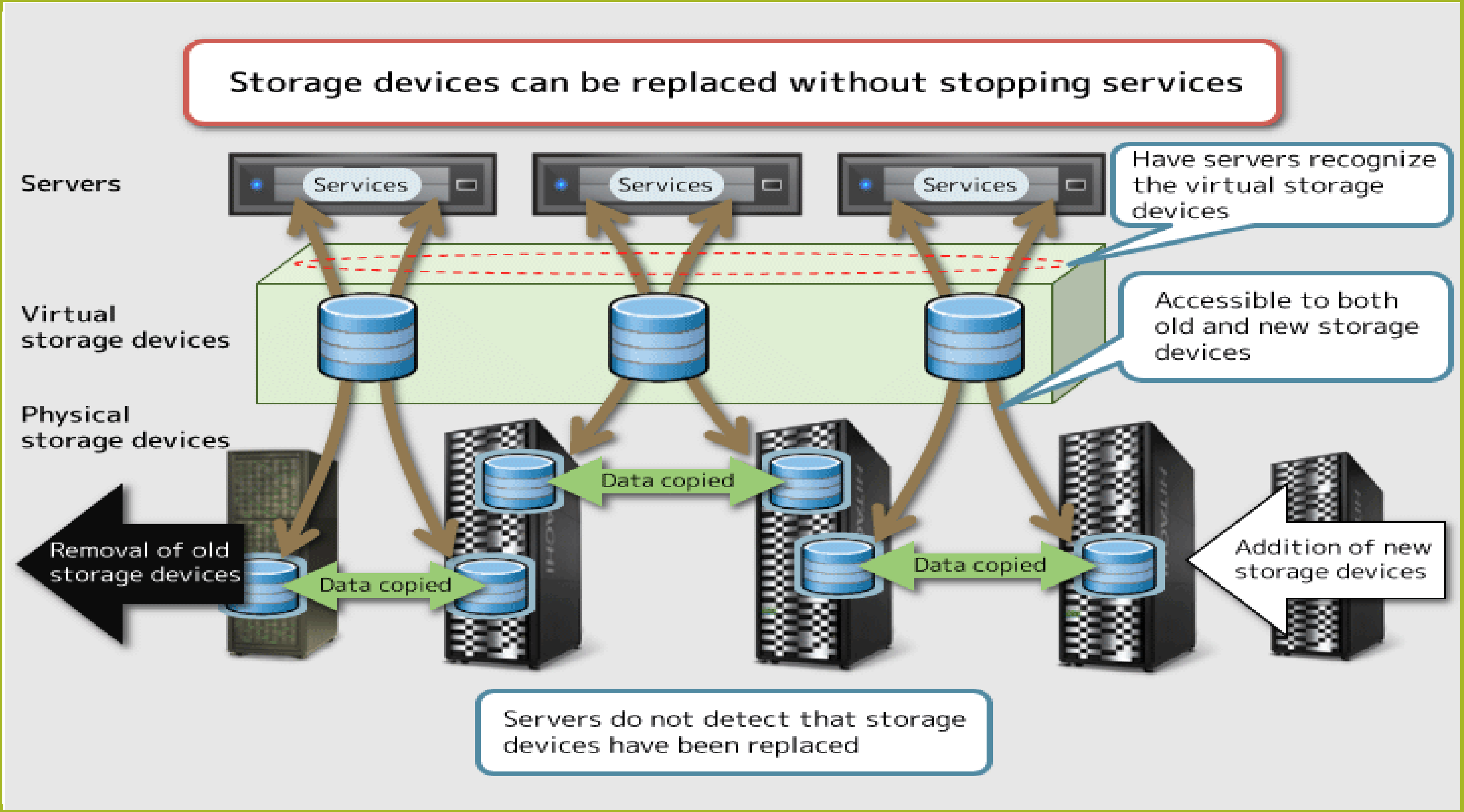
How it works?

Simplified Management:

Storage virtualization simplifies storage management by *centralizing the control and administration of the storage infrastructure*. Administrators can perform tasks like *capacity allocation, performance optimization, and data protection* from a single management interface, regardless of the underlying physical storage devices.

Flexibility and Scalability:

Storage virtualization offers flexibility and scalability by decoupling the logical storage units from the physical hardware. It allows for *easy expansion and addition of storage capacity* without disrupting the applications or systems relying on the virtual volumes.



Storage devices can be replaced without stopping services
Servers
Services
Services
Services
Have servers recognize the virtual storage devices
Virtual storage devices
Accessible to both old and new storage devices
Physical storage devices
Data copied
Removal of old storage devices
Data copied
Data copied
Addition of new storage devices
Servers do not detect that storage devices have been replaced

Types of Storage Virtualization

1. Block Level Storage

Block level storage, or block storage, is storage used for structured data and is commonly deployed in Storage Area Network (SAN) systems.

It uses Internet Small Computer Systems Interface (iSCSI) and Fibre Channel (FC) protocols.

SAN is a high-speed, dedicated network of servers and shared storage devices and provides access to consolidated, block level data storage.

Types of Storage Virtualization

1. Block Level Storage

Block-level virtualization is a method that operates at the block device layer, which is the level that reads and writes the data blocks on a storage device.

Block-level virtualization allows *multiple block devices to be aggregated and presented as a single logical unit*, regardless of their physical characteristics, such as capacity, speed, or vendor.

This can *enhance the performance, availability, and utilization of storage resources*

Types of Storage Virtualization

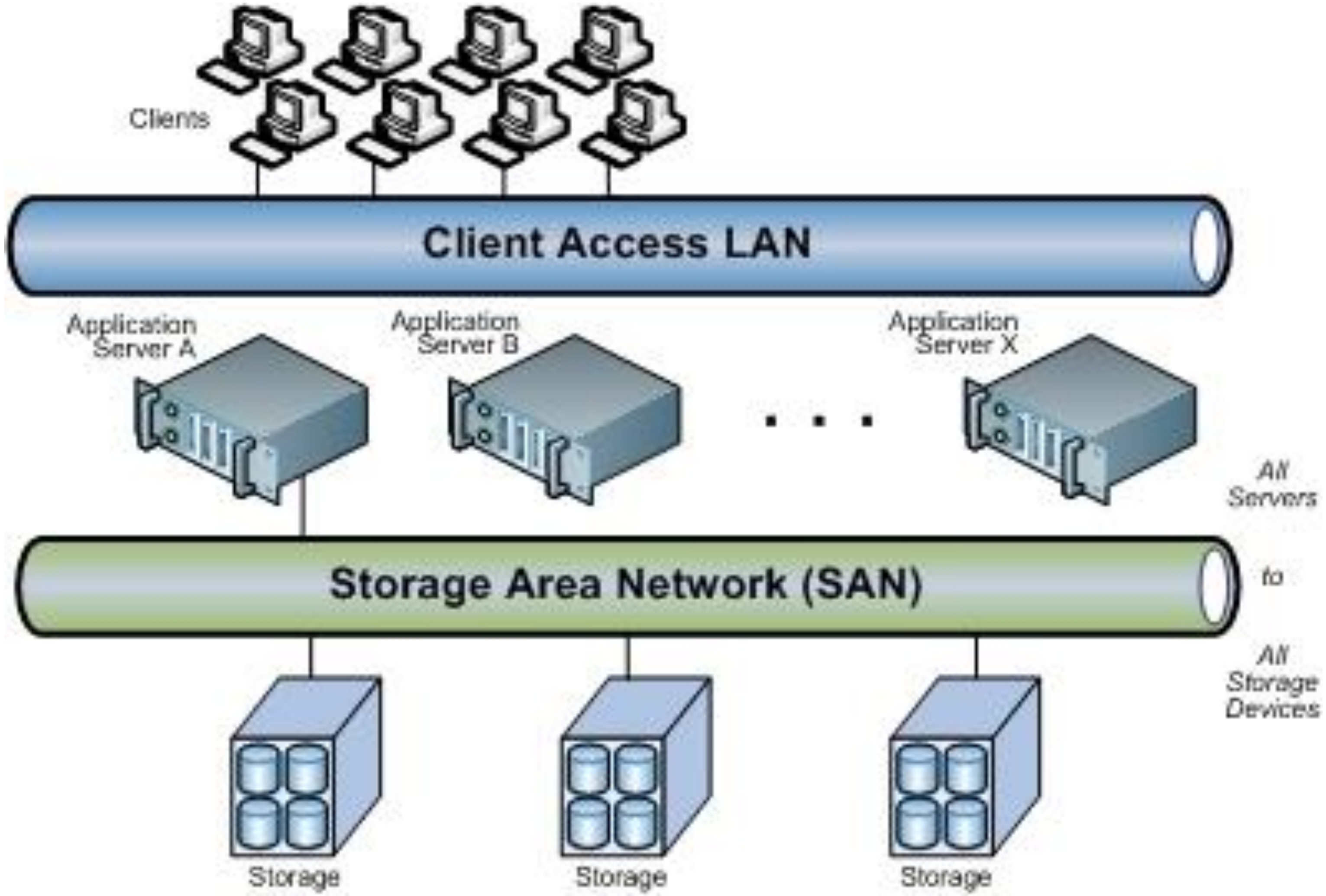
1. Block Level Storage

Block storage uses blocks, which are a *set sequence of bytes*, to store structured workloads.

Each block is *assigned a unique hash value which functions as an address*. In block storage, the data is stored without any metadata e.g. data format, type, ownership, etc.

The ability to store data in blocks delivers structured workloads such as databases, applications, etc.

Storage Area Network



Types of Storage Virtualization

2. File Level Storage

- File level storage, or file storage, is storage used for unstructured data and is commonly deployed in Network Attached Storage (NAS) systems.
- Network attached storage commonly called as NAS and is an IP-based file sharing device which is attached to a local area network (LAN)
- It uses Network File System (NFS) for Linux, and Common Internet File System (CIFS) or Server Message Block (SMB) protocols for Windows.
- File storage, as opposed to block storage, *stores data in a hierarchical architecture*; as such that the data and its metadata are stored as is – in the form of files and folders.

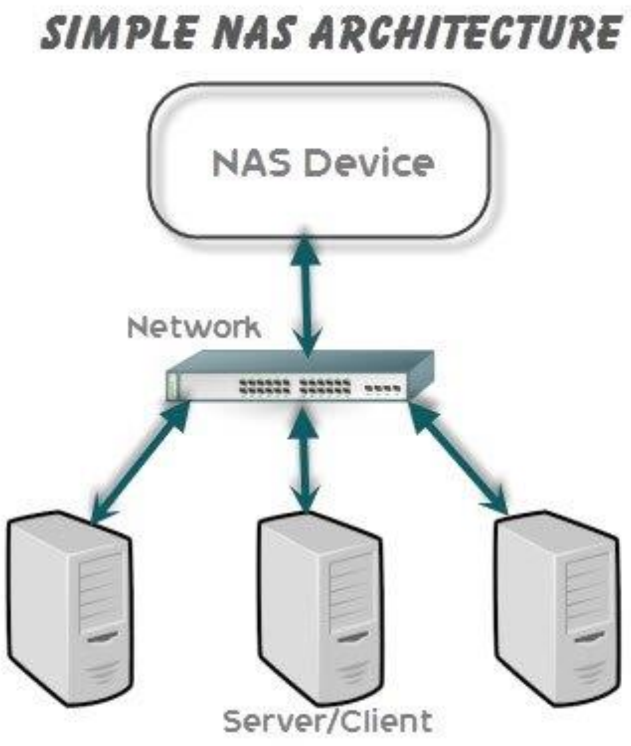
Types of Storage Virtualization

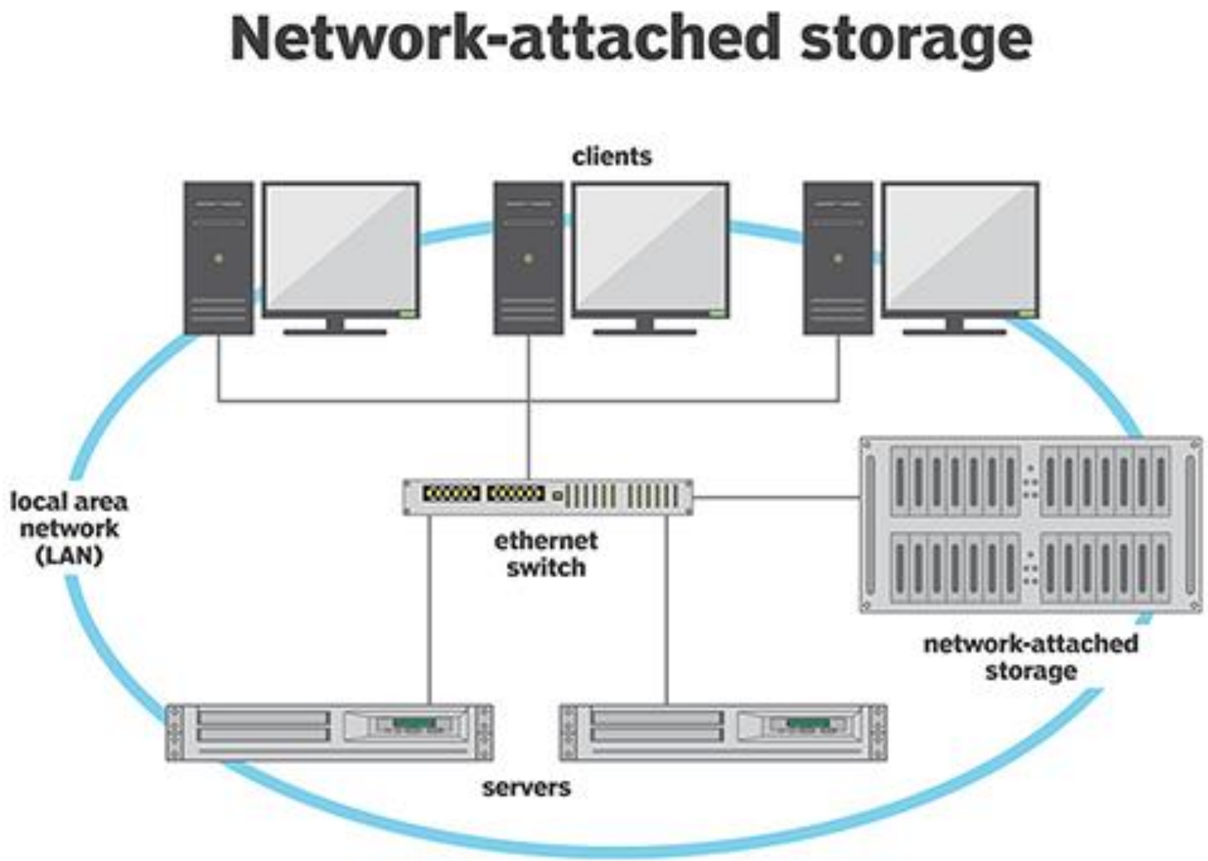
2. File Level Storage

- File-level virtualization is a method that operates at the file system layer, which is the level that organizes and manages the files and directories on a storage device.
- File-level virtualization allows multiple file systems to be pooled together and accessed as a single namespace, regardless of their physical location, size, or format.
- This can simplify the administration and migration of files, as well as provide load balancing and fault tolerance.

Types of Storage Virtualization

2. File Level Storage





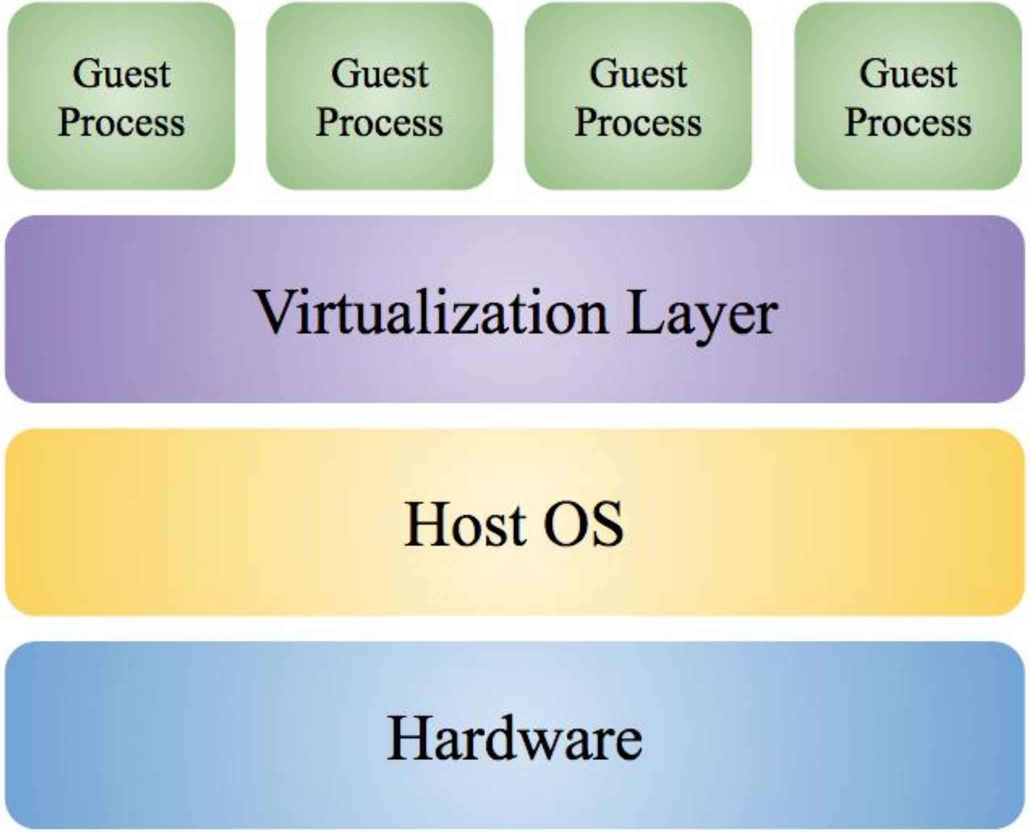
Types of Storage Virtualization

2. File Level Storage

Drawbacks

- overhead of maintaining the metadata and the mapping of files,
- the potential inconsistency of file attributes and permissions across different file systems, and
- the lack of granularity and efficiency for applications that require low-level access to the data blocks

OS based Virtualization



OS-Level Virtualization

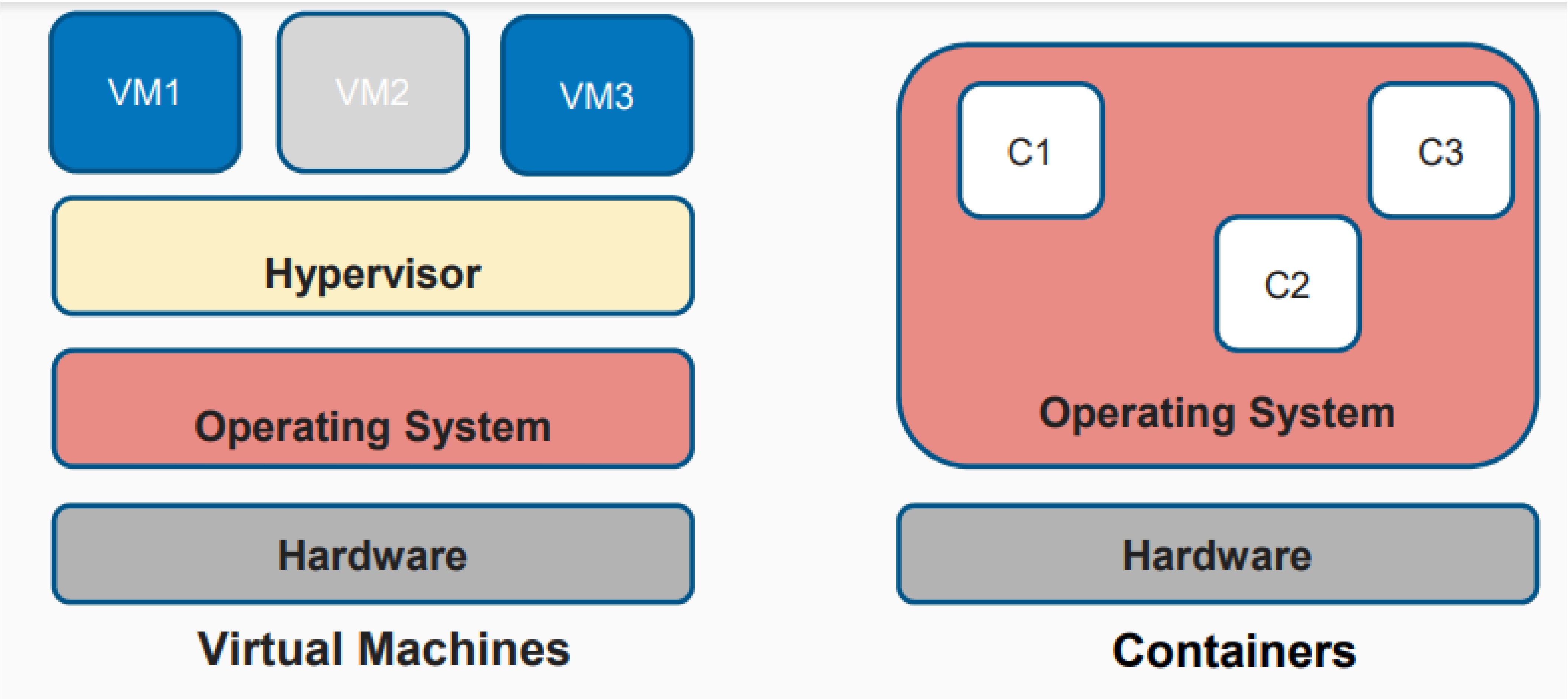
OS based Virtualization

- OS-level virtualization, also known as *containerization*, is a type of virtualization technology that allows *multiple isolated user-space instances*, called containers, to run on a single host operating system.
- It is a virtual execution environment that can be forked instantly from the base operating environment.
- Here the kernel of an OS allows more than one isolated user-space instances to exist. Such instances are called *containers/software containers or virtualization engines*.
- These containers share the same OS kernel but have their own file systems, processes, and network interfaces, providing isolation and security similar to virtual machines but with less overhead.

OS based Virtualization - Features

1. **Isolation:** Containers operate in isolated environments, ensuring that the processes inside a container do not interfere with those in other containers or the host system.
2. **Efficiency:** Since containers share the host OS kernel, they are generally more lightweight and efficient compared to traditional virtual machines which require separate guest OS instances.
3. **Portability:** Containers can be easily moved between different environments (e.g., development, testing, production) without changes to the code
4. **Scalability:** Containers can be quickly scaled up or down, making them ideal for micro-services and cloud-native applications.
5. **Resource Allocation**: Containers can have resource limits set on CPU, memory, and other system resources to ensure fair distribution and avoid overconsumption.

How Virtual Machines are different from Containers



OS based Virtualization - Benefits

Reduced Overhead: Lower resource usage compared to traditional VMs as containers share the host OS kernel.

Fast Deployment: Quick startup and shutdown times, enabling rapid deployment and scaling.

Improved DevOps: Containers facilitate DevOps practices by ensuring consistency across development, testing, and production environments.

OS based Virtualization - Challenges

1. **Security:** Although containers provide isolation, they share the host OS kernel, which can pose security risks if not properly managed.
2. **Networking:** Managing network configurations for containers can be complex, especially in large-scale deployments.
3. **Storage:** Persistent storage for containers can be challenging and often requires specialized solutions.

Popular OS-level Virtualization Technologies

1. **Docker:** One of the most popular containerization platforms that simplifies the creation, deployment, and management of containers.
2. **Kubernetes:** An orchestration platform for managing and scaling containerized applications.
3. **LXC (Linux Containers):** A low-level Linux container runtime that provides a more lightweight virtualization method.
4. **Podman:** An alternative to Docker that allows you to manage containers without requiring a daemon.

Cost of Virtualization

The direct cost of virtualization is based on three factors:

1. Hardware expenses
2. Software licensing fees
3. Implementation and deployment cost

The indirect cost of virtualization is based on three factors:

1. Training and Personnel costs
2. Maintenance, support and upgrades
3. Energy consumption